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Assignment 6A

Aim

Write a program to imlement insert, delete and taversal on a singly linked list

Theory

A linked list is an ordered collection of data where each element (generally called nodes) contains the location of the next element in the list.

Each node has two parts - ~~one~~ the data part and the link which connects it to the next one which is usually called next

Insertion.

In a linked list mean adding a new node to the list A new node is created, its data is stored and its address is connected to the next node. The previous node's link is updated to point to this new node.

Deletion

Means removing a node from the linked list. The link of the previous node is changed so that it skips the node to be deleted. Then the memory of the deleted node is freed.

Traversal

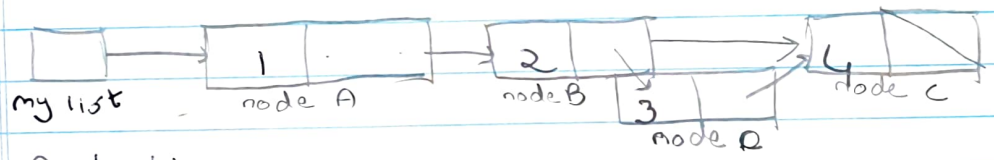
It means visiting each node of the linked list

one by one. We start from the first node (head) and follow the links using help pointer until we reach NULL. This is usually done to display or search elements in the list.

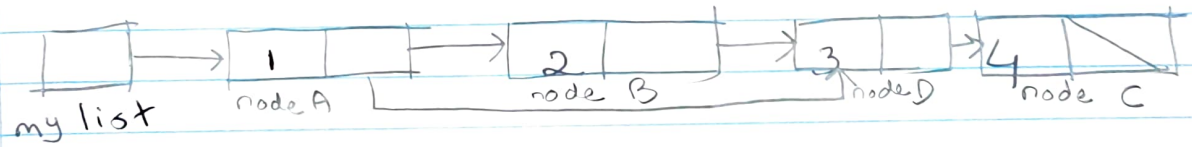
Time Complexity
Insertion & Deletion - O(1)
Traversal - O(n)

Conclusion
The concept of linked lists is applied to evaluate problems in a variety of applications

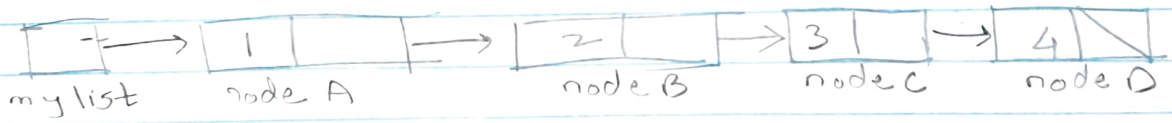
e.g. Insertion



Deletion



Traversal



```
#include <stdio.h>
#include <stdlib.h>

struct ll_node {
   int data;
   struct ll_node *next;
};

struct ll_node* del(struct ll_node* head, int val);
struct ll_node* insert(struct ll_node *head, int val);
void display(struct ll_node *head);

void main() {

   struct ll_node *myList = NULL;
   int choice, value;

   do {
      printf("\n------ MENU ------\n");
      printf("1-Add Element\n");
      printf("2-Remove Element\n");
      printf("3-Show Linked List\n");
      printf("4-Quit\n");
      printf("Select option: ");
      scanf("%d", &choice);

      switch(choice) {
```

```
        case 1:
            printf("Input number to insert: ");
            scanf("%d", &value);
            myList = insert(myList, value);
            break;
        case 2:
            printf("Input number to remove: ");
            scanf("%d", &value);
            myList = del(myList, value);
            break;

        case 3:
            display(myList);
            break;

        case 4:
            printf("Program terminated.\n");
            break;

        default:
            printf("Wrong choice! Try again.\n");
    }

  } while(choice != 4);
}

struct ll_node* del(struct ll_node* head, int val) {
  struct ll_node *temp = head;
```

```
    if (temp != NULL){
        struct ll_node *prev = NULL;
        while (temp != NULL){
            if(temp->data == val){
                struct ll_node *rem = temp;
                if (prev == NULL){
                    head = temp->next;
                } else {
                    prev->next = temp->next;
                }
                free(rem);
                return head;
            }
            prev = temp;
            temp = temp->next;
        }
        printf("ELEMENT NOT FOUND");
        return head;
    }
    printf("LIST IS EMPTY");
    return head;
}

struct ll_node* insert(struct ll_node *head, int val) {

    struct ll_node *temp = head;

    struct ll_node *newNode = (struct ll_node*) malloc(sizeof(struct ll_node));
```

```
    newNode->data = val;
    newNode->next = NULL;

    if (head == NULL || head->data > val) {
        newNode->next = head;
        return newNode;
    }

    while (temp->next != NULL && temp->next->data < val) {
        temp = temp->next;
    }

    newNode->next = temp->next;
    temp->next = newNode;

    return head;
}

void display(struct ll_node *head) {

    printf("\nLinked List Elements:\n");

    while (head != NULL) {
        printf("%d  ", head->data);
        head = head->next;
    }

    printf("\n");
```

}

```
------ MENU ------
1-Add Element
2-Remove Element
3-Show Linked List
4-Quit
Select option: 1
Input number to insert: 10

------ MENU ------
1-Add Element
2-Remove Element
3-Show Linked List
4-Quit
Select option: 1
Input number to insert: 30

------ MENU ------
1-Add Element
2-Remove Element
3-Show Linked List
4-Quit
Select option: 1
Input number to insert: 20

------ MENU ------
1-Add Element
2-Remove Element
3-Show Linked List
4-Quit
Select option: 3

Linked List Elements:
10  20  30
```

```
------ MENU ------
1-Add Element
2-Remove Element
3-Show Linked List
4-Quit
Select option: 2
Input number to remove: 10

------ MENU ------
1-Add Element
2-Remove Element
3-Show Linked List
4-Quit
Select option: 3

Linked List Elements:
20  30
```